

(a) transport-masked Feshbach map

microscopic Hamiltonian

PHP	PHQ
QHP	QHQ

ice P

spinons Q

fold out Q
 $F(z)$ on P

ice states sorted by transport $\mathcal{T}$

	×	
×		×
		×

red: torus-changing matrix elements

keep $\mathcal{T}_a = \mathcal{T}_b$

$\Delta[F(z)]$

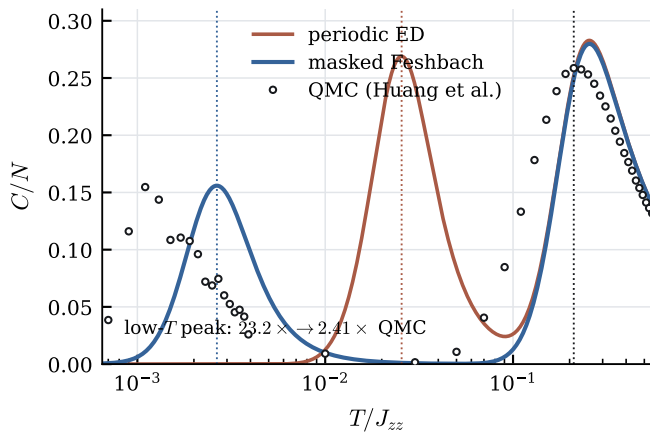
What the mask means

$\mathcal{T}_a = \mathcal{T}_b$: a local loop closes on the covering lattice $\rightarrow$ keep

$\mathcal{T}_a \neq \mathcal{T}_b$: a path closes only through a periodic image $\rightarrow$ remove

Self-consistent roots of the remaining blocks define the finite-volume theory.

(b) sign-free benchmark, $J_{\pm}/J_{zz} = +0.046$



(c) false $J_{\pm}^2$ scale removed

